Adesh Grewal MProfE
Mechatronics Engineer

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LinkedIn: https://www.linkedin.com/in/grewaladesh13|| **Portfolio**: https://ag-013.github.io/

Summary:
Robotics Engineer with hands-on experience in robots, Student Formula and agricultural machinery. With practical experience in field deployment of robotics.
Interested in Agri-robotics and control systems.
Work Rights: Full work rights in Australia (Graduate 485)

Skills:
Mechanical & Simulation
- SolidWorks (CAD) and ANSYS - Design and component structural simulation.
- 3D-printing, Rapid Prototyping, Component Manufacturing

Embedded & Hardware
- Embedded C/C++, STM32, Motor Control, Hardware Integration, Ethernet,

Robotics & Autonomy
- ROS / ROS2, Robot Exploration, Path Planning, Multi-Robot Collaboration

Control & Systems
- Autonomous Control, System Design, Algorithm Development

Programming & Simulation
- C/C++, Python, MATLAB, Gazebo

Experience:

Robotics Field Engineer — Feb 2026 - Present
EVS Broadcasting Equipment, Sydney
- Delivered remote and on-site support, Debugging and issue resolution for hardware, software and electronic issues.
- Managing customers and coordinating on-site visits for system health checks.
- Performed system commissioning, troubleshooting, diagnostics, and repairs to maximize equipment uptime.

Casual Academic - Mechatronics — July 2025 - December 2025
University of Sydney, Sydney
- Facilitated 3 weekly mechatronics lab sessions, covering circuits and embedded systems.
- Tutored 120+ undergraduate students, improving lab and conceptual understanding.

Research Engineer — October 2023 - November 2024
University of Technology Sydney, Sydney
- Conducted vibrational analysis on flower petals using Siemens Sim center to study the relationship between bee-induced vibrations and pollen release.
- Evaluated 15+ acoustic insulation material combinations, quantifying noise reduction performance for curtain-based acoustic insulation.

Graduate Engineer — July 2022 - June 2023
Mahindra & Mahindra, Chandigarh, India
- Supervised a new shot-blasting production line, ensuring stable and efficient operation.
- Optimised shot usage and blasting cycle times for tractor gearbox and differential.
- Performed root cause analysis with Quality Assurance team to resolve surface defects.

Projects:
Robotic Arm–Equipped Guide Dog Platform
- Designed a 3-DOF robotic arm connected with a legged robot to assist visually impaired users.

- Implemented feedback control, facilitating safe human-robot collaboration.
- Integrated and 3D-printed mechanical components with ROS, Gazebo, and microcontrollers.

Autonomous Navigation in Unstructured Environments

- Applied Frontier-based exploration for autonomous navigation in unknown environments.
- Integrated YOLO-based object detection into a ROS navigation stack for obstacle identification.

Multi-Robot Collaboration & Surveying System

- Configured multi-robot coordination via ROS2.
- Deployed across platforms including quadcopters, Ackermann-steered, and skid-steer robots.
- Enabled coordinated navigation and area surveying.

Vision-Based Robotic Manipulation

- Implemented image-guided pick-and-place with depth cameras, and eye-to-hand calibration.
- Coupled with computer vision and motion control for reliable object grasping.

Autonomous Fixed-Wing UAV for Surveillance

- Engineered a fixed-wing UAV for crop and land surveying and surveillance missions.
- Deployed with horizon stabilization and mission planning and execution capability

Student Motorsports

- UTS Autonomous Mechatronics Lead for team. Represented at SXSW 2024.
- Designed a switchable pedal box for autonomous and manual operation
- Led Ackermann steering across three go-kart platforms, optimizing turn radius and grip.

Publications:

- Singh, R., Grewal, A., Singh, A.P. *et al.* 3D printed sensor for online condition monitoring of energy storage devices. *Sādhanā* **47**, 212 (2022). https://doi.org/10.1007/s12046-022-01992-2

Education:

Master of Engineering **August 2023 - July 2025**

University of Technology Sydney, Sydney, Australia

- **GPA: 5.56 / 7**
- **Graduate Project:** Robotic Arm-Equipped Guide Dog for the Visually Impaired. ● Key Coursework: Robot Exploration & Navigation, Autonomous Control, Real-Time Simulation

Bachelor of Mechanical Engineering **August 2018 - July 2022**

University Institute of Engineering and Technology, Chandigarh, India

- **CGPA: 7.25/10**
- **Graduate Project:** Autonomous Fixed-Wing UAV for Surveillance

References:

Dr. Donald Dansereau, University of Sydney
Dr. Marc Carmichael, UTS
Dr. Can Nerse, UTS Tech Labs